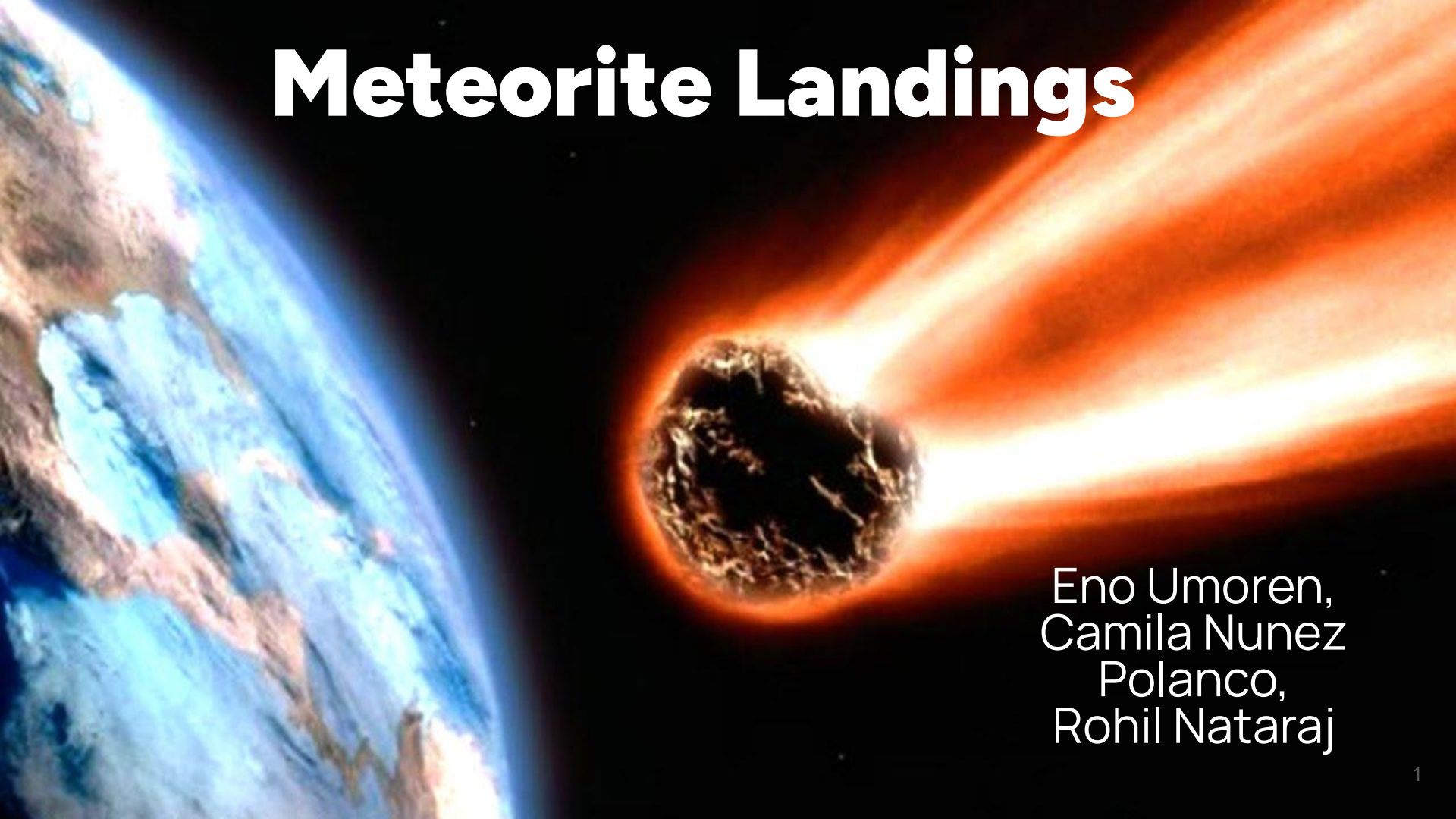
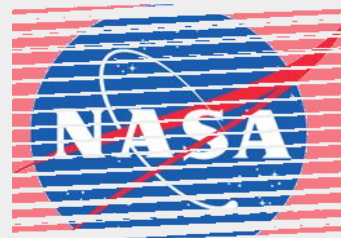


Meteorite Landings

A dramatic image of a meteorite streaking across the sky towards Earth. The Earth's curved horizon is visible on the left, showing blue oceans and brown landmasses. A large, dark, rocky meteorite is in the center, surrounded by a bright, glowing orange and yellow fireball. A long, bright streak of light extends from the meteorite towards the right side of the frame.

Eno Umoren,
Camila Nunez
Polanco,
Rohil Nataraj

Data



- *Meteorites Landings* from NASA Open Data
- Composed by The Meteoritical Society
- Contains **34,000+ recorded meteorites** observed or discovered on Earth
- Covers meteorite events from **860 CE to 2013**
- **Unit of analysis:** one observation = one meteorite event
- Meteorites give insight into **solar system formation and composition**
- Dataset allows exploration of **patterns in meteorite discovery vs. observed falls**

Pre-Processing

- Incorrect Year Values
 - Some dates (<860 CE or > 2016) were improperly converted in NASA dataset
- Invalid Location Coordinates
 - Some entries recorded at (0°N, 0°E), treated as missing (NA) and removed
- Final dataset: ~ **31,000 recorded meteorites**
- Many variables were:
 - **Continuous** (e.g., mass, year)
 - Or had **too many unique values**

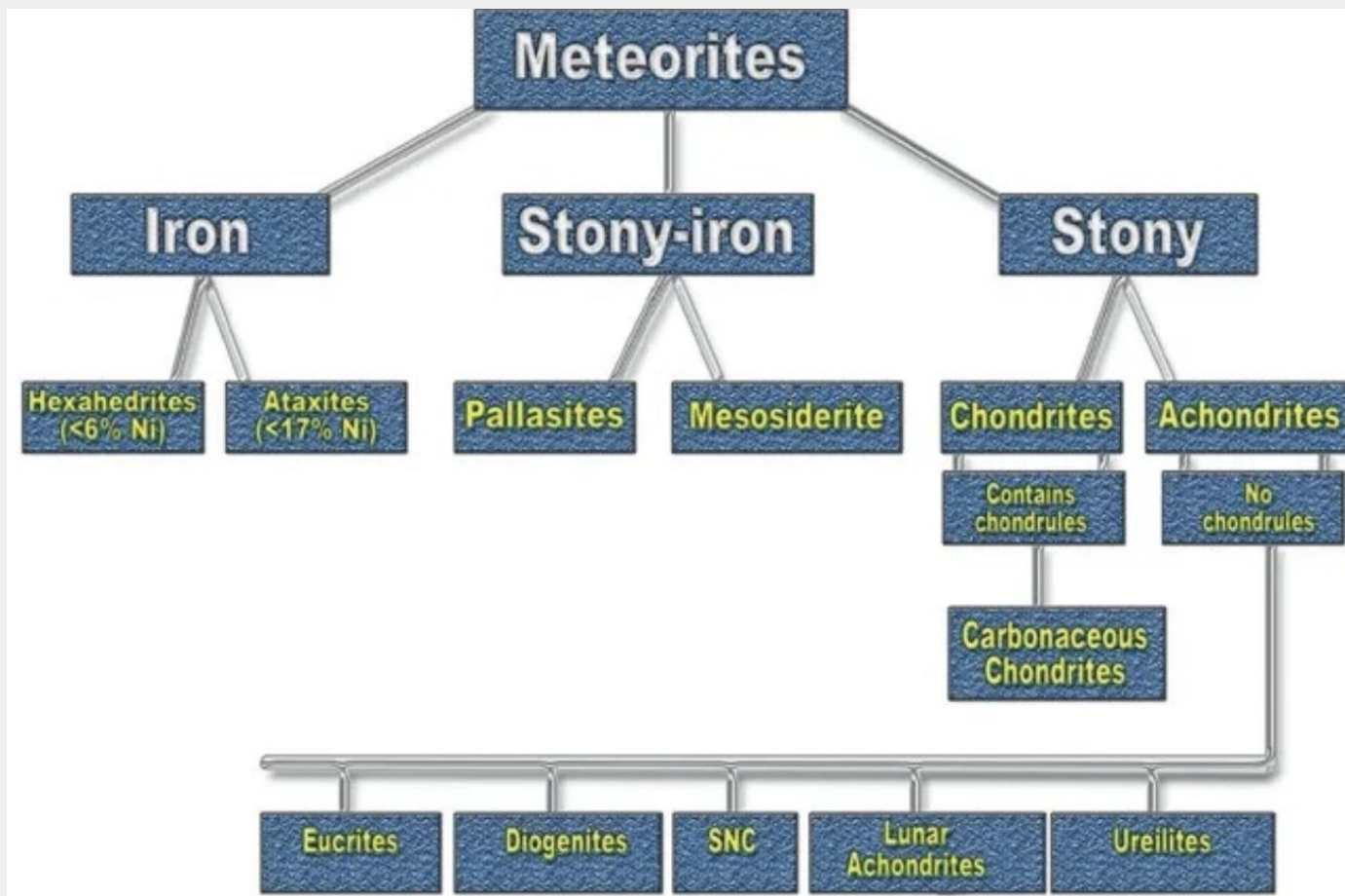
Research Question & Method

Question: Which factors (composition, mass, year, location) are associated with the probability that a meteorite is classified as “**Fell**” vs “**Found**”?

- Chi-square Test
- Logistic Regression (binary outcome: Fell vs Found)
- Odds Ratio

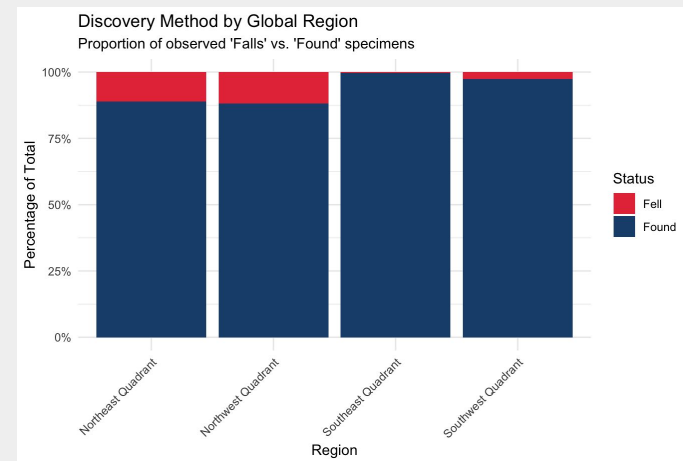
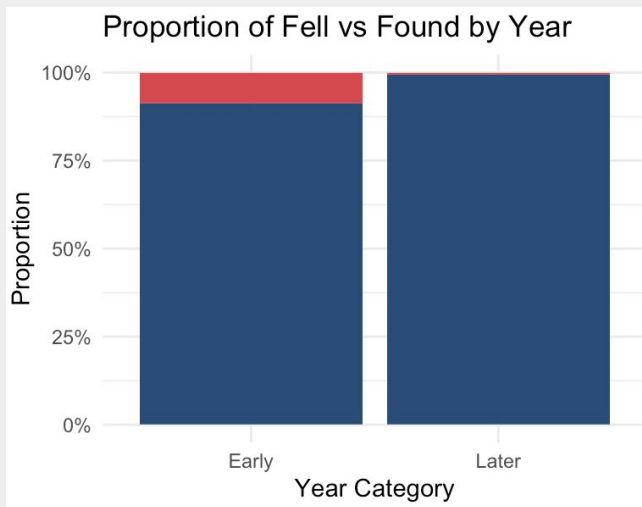
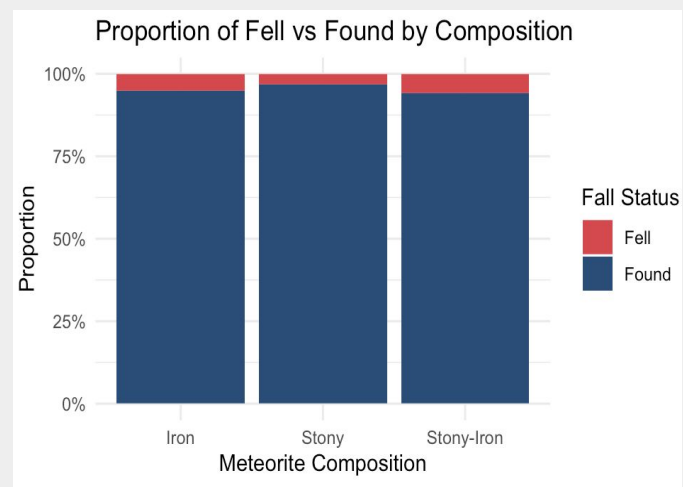
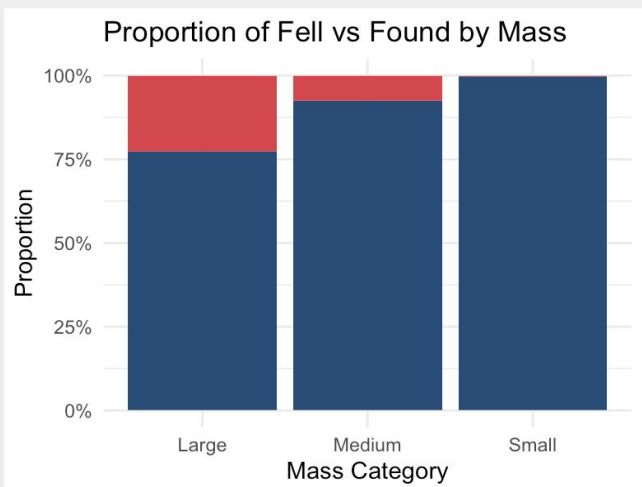
Transformed Key Variables

Variable	Description
Fell vs Found (categorical)	Observed falling vs discovered later
Composition	Stone, Iron, Stony-Iron
Mass (grams)	Small, Medium, Large
Year discovered	Early (860 CE - 1986), Later (1987 - 2013)
Geographic Location	Northeast, Northwest, Southeast, Southwest quadrant

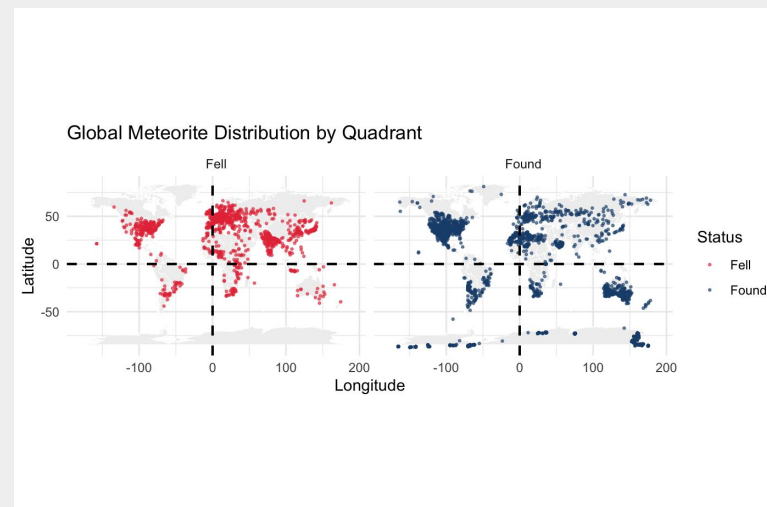
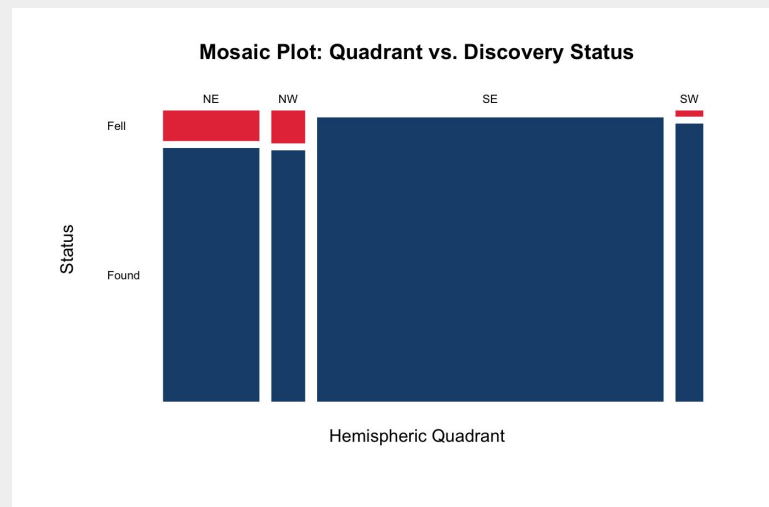
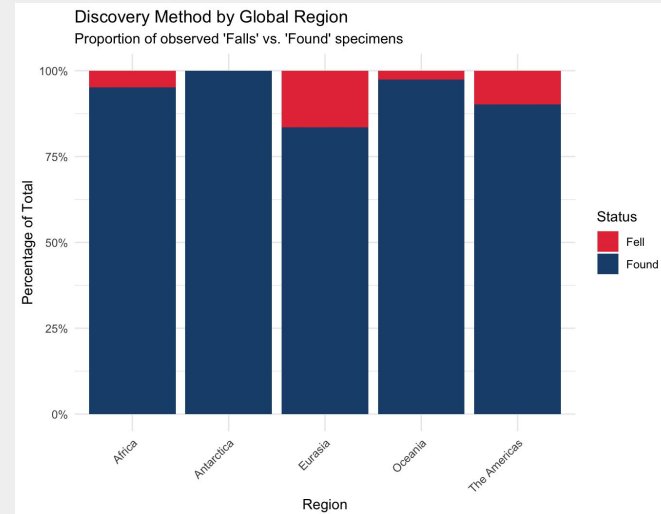
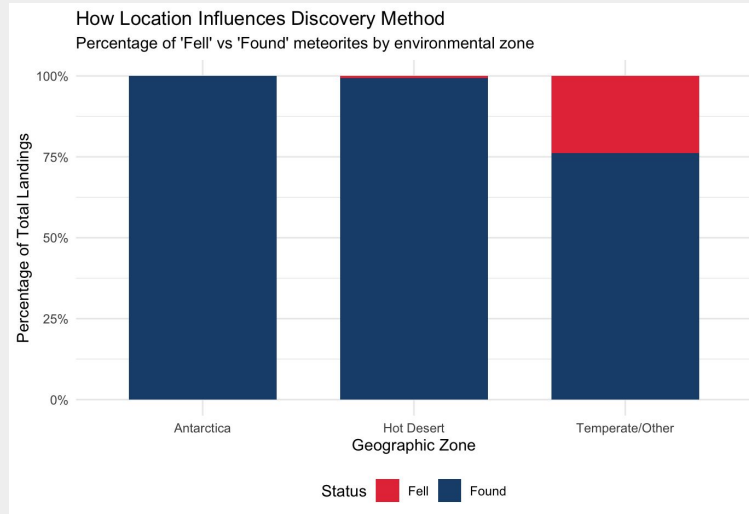


(Gould,
2021)

EDA



EDA



Chi-Square Tests of Association with “Fell” Classification

- All expected counts > 5
 - Chi-square assumptions satisfied

Predictor	χ^2	df	P-value
Mass	2554.4	2	< 0.001
Composition	14.12	2	$p = 0.00086$
Year	1433.7	1	< 0.001
Region	2222.2	3	< 0.001

- All predictors are significantly associated with classification, with **mass**, **composition**, **year**, **region**

Statistical Assumptions

Chi-Square Test Assumptions

- Categorical variables
- Expected cell counts > 5
- Large sample size (~31,000 observations)

Logistic Regression Assumptions

- Binary outcome variable (Fell vs Found)
- Independent observations
- Limited multicollinearity among predictors
- Large sample size
- No perfect separation of classes
- Log-odds modeled as a linear combination of predictors

Training vs Testing Data

- Dataset randomly split into:
 - 70% training data
 - 30% testing data
- Training data used to:
 - Fit logistic regression models
 - Estimate coefficients and odds ratios
- Testing data used to:
 - Evaluate predictive performance
 - Generate ROC curves and confusion matrices

Baseline Model

Model Setup

- Reference category: **Found**
- Outcome: probability of **Fell (observed falling)**

Fall ~ 1	Coefficient	P-value	P(Fell)	OR
Intercept	-3.3665	<2e-16	3.3%	0.0345

- Null Deviance: **6531.7** (df = 22335)
- AIC: **6533.7**

Model Fall ~ Mass

Model Setup

- Reference category: **Large**
- Outcome: probability of **Fell (observed falling)**

Fall ~ Mass	Coefficient	P-value	P(Fell)	OR
Intercept (Large)	-1.28036	<2e-16	21.7%	0.278
Medium	-1.21845	<2e-16	7.5%	0.296
Small	-4.39080	<2e-16	0.34%	0.012

- Residual Deviance: **6531.6**
- AIC: **6535.6**

Model Fall ~ Composition

Model Setup

- Reference category: **Iron**
- Outcome: probability of **Fell (observed falling)**

Fall ~ Composition	Coefficient	P-value	P(Fell)	OR
Intercept (Iron)	-2.9973	<2e-16***	4.75%	0.05
Stony	-0.3888	0.0357*	3.27%	0.679
Stony-Iron	0.3146	0.4406	6.4%	1.37

- Residual Deviance: **6531.7** (df = 22335)
- AIC: **6530.8**

Model Fall~Region

Pearson's Chi-squared test:

Data:

Quadrant	Fell	Found
NE	676	5436
NW	260	1930
SE	82	21711
SW	46	1767

X-squared = 2222.2, df = 3, P-value < 2.2e-16

Standardized Residuals:

Quadrant	Fell	Found
NE	33.08	-6.14
NW	21.88	-4.06
SE	-23.92	4.44
SW	-1.86	0.35

Model Fell~Region

Data: Meteor_quadrants (Trained Data)

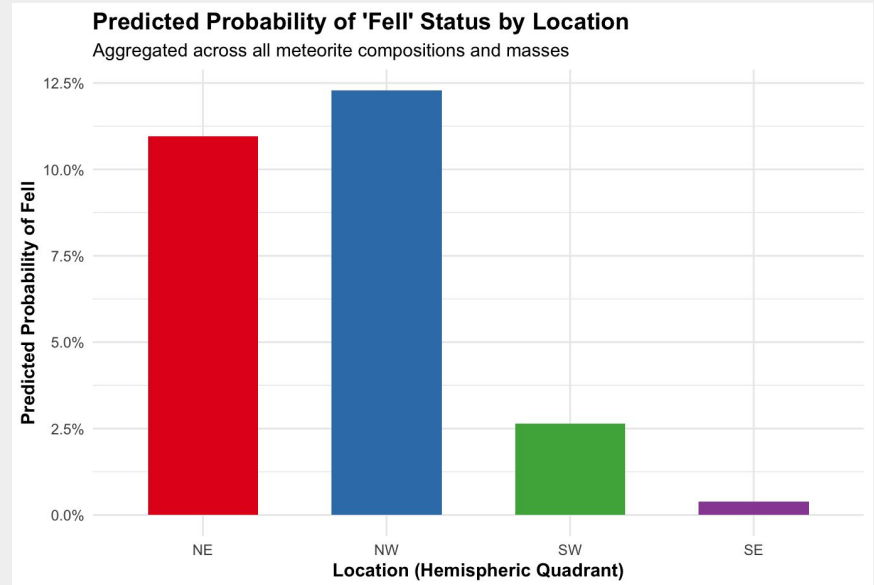
Reference: **SE quadrant**

Quadrant	Probability (%)	Estimate
SE (intercept)	0.38	-5.5720
NE	10.96	3.4776
NW	12.28	3.6059
SW	2.65	1.9677

Null Deviance: **6531.7**

Residual deviance: **5146.7**

AIC: **5146.7**



Model Fell ~ Year

Coefficients	Estimate	P-value	Odds
Intercept	-2.65172	<2e-16	0.070
Later	-2.27194	<2e-16	0.103

AIC: 5844, Baseline: Early

Null deviance: 6428.2 on 22302 degrees of freedom

Residual deviance: 5840.0 on 22301 degrees of freedom

Mass + Region

- Reference: **Large + Northeast**
- Residual Deviance: **4739.0**
- AIC: **4751**

Fall ~ Mass + Region	Estimate	P-value	OR
Intercept	-0.9038	<2e-16	0.405
Medium	-1.0505	<2e-16	0.349
Small	-3.0627	<2e-16	0.047
Northwest	-0.1784	0.0781	0.837
Southeast	-2.3924	<2e-16	0.091
Southwest	-1.1054	<2e-16	0.331

Full model

AIC: 6430.2 **Baseline:** Large + Iron + Northeast + Early

Null deviance: 6428.2 **Residual deviance:** 6428.2

Coefficients	Estimate	P-value	Odds
Intercept	-1.0897	1.16e-08	0.336
Medium	-0.5683	2.34e-05	0.566
Small	-2.3289	<2e-16	0.097
Stony	2.5603	<2e-16	12.9340
Stony-Iron	0.5911	0.279	1.806

Coefficients	Estimate	P-value	Odds
NW	-1.8929	<2e-16	0.151
SE	-4.7794	<2e-16	0.0084
SW	-1.5660	4.96e-12	0.209
Later	-4.3782	<2e-16	0.0125

Interaction Model

- Tested whether the effect of meteorite mass on “Fell” vs “Found” changes across geographic regions
- Reference: **Large, Northeast**
- Mass, region, composition, and years were all significant predictors
- Interaction term showed mass **does not** have the same effect world wide

```
glm(formula = fall ~ mass_cat * quadrant + class_group + year_cat2,  
     family = "binomial", data = train_data)
```

```
Null deviance: 6531.7  on 22335  degrees of freedom  
Residual deviance: 2987.3  on 22321  degrees of freedom  
AIC: 3017.3
```

Interaction Model Results

Medium × Northwest Quadrant (+1.1879):

- Medium meteorites in the Northwest quadrant had about **3.3 times higher odds** of being classified as “Fell” compared to what would be expected from the individual effects alone

Small × Northwest Quadrant (+1.0153):

- Small meteorites in the Northwest quadrant had approximately **2.8 times higher odds** of being classified as “Fell”

Medium × Southeast Quadrant (−1.1361):

- Medium meteorites in the Southeast quadrant had only about **0.32 times the odds** (roughly a **68% decrease in odds**) of being classified as “Fell”

Small × Southeast Quadrant (−4.5338):

- Small meteorites in the Southeast quadrant had only about **1% of the odds** of being classified as “Fell,” representing an extremely strong negative interaction effect

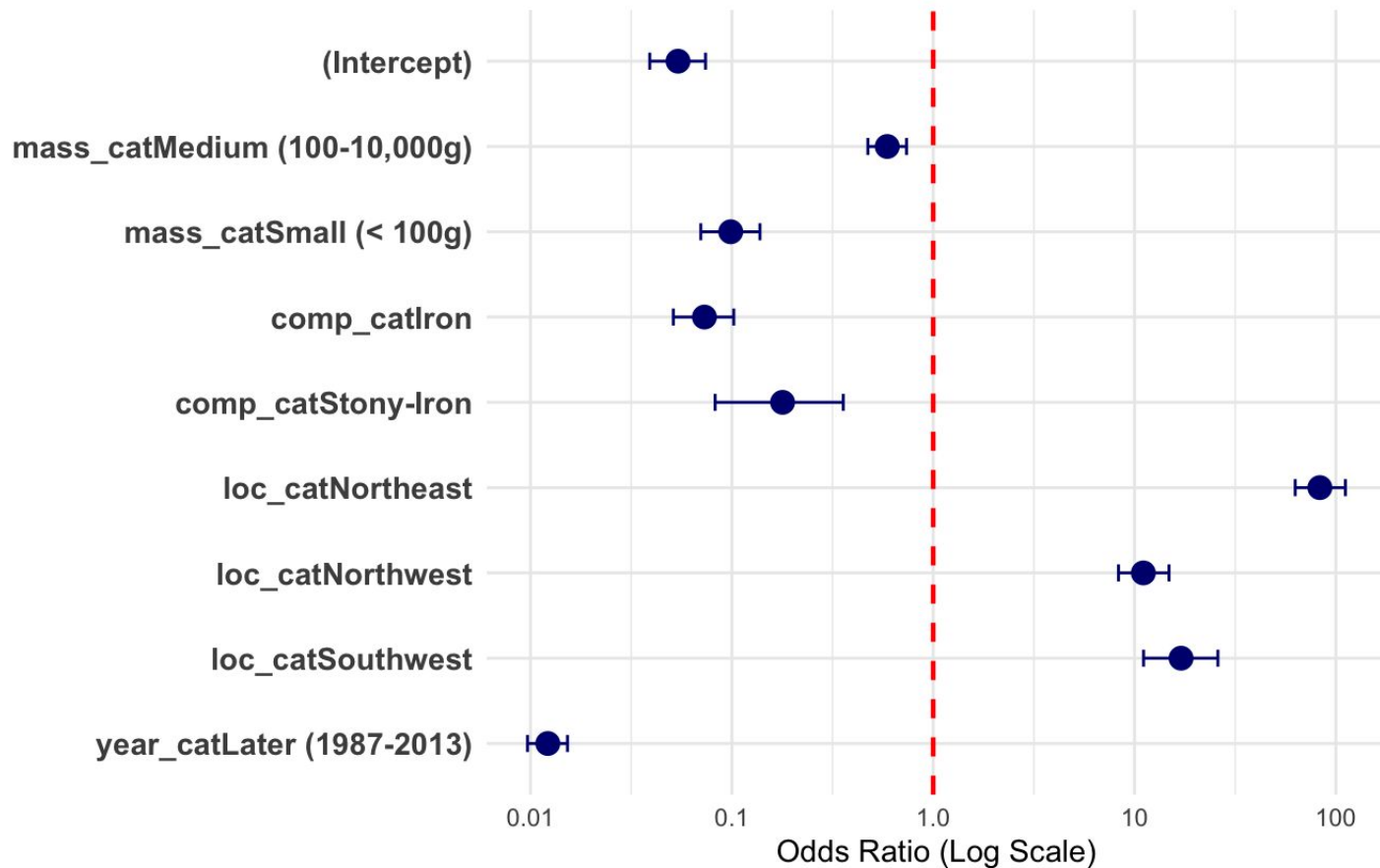
Odd Ratios

Reference: **Large**, **Stony**, **SE quadrant**, **Early**

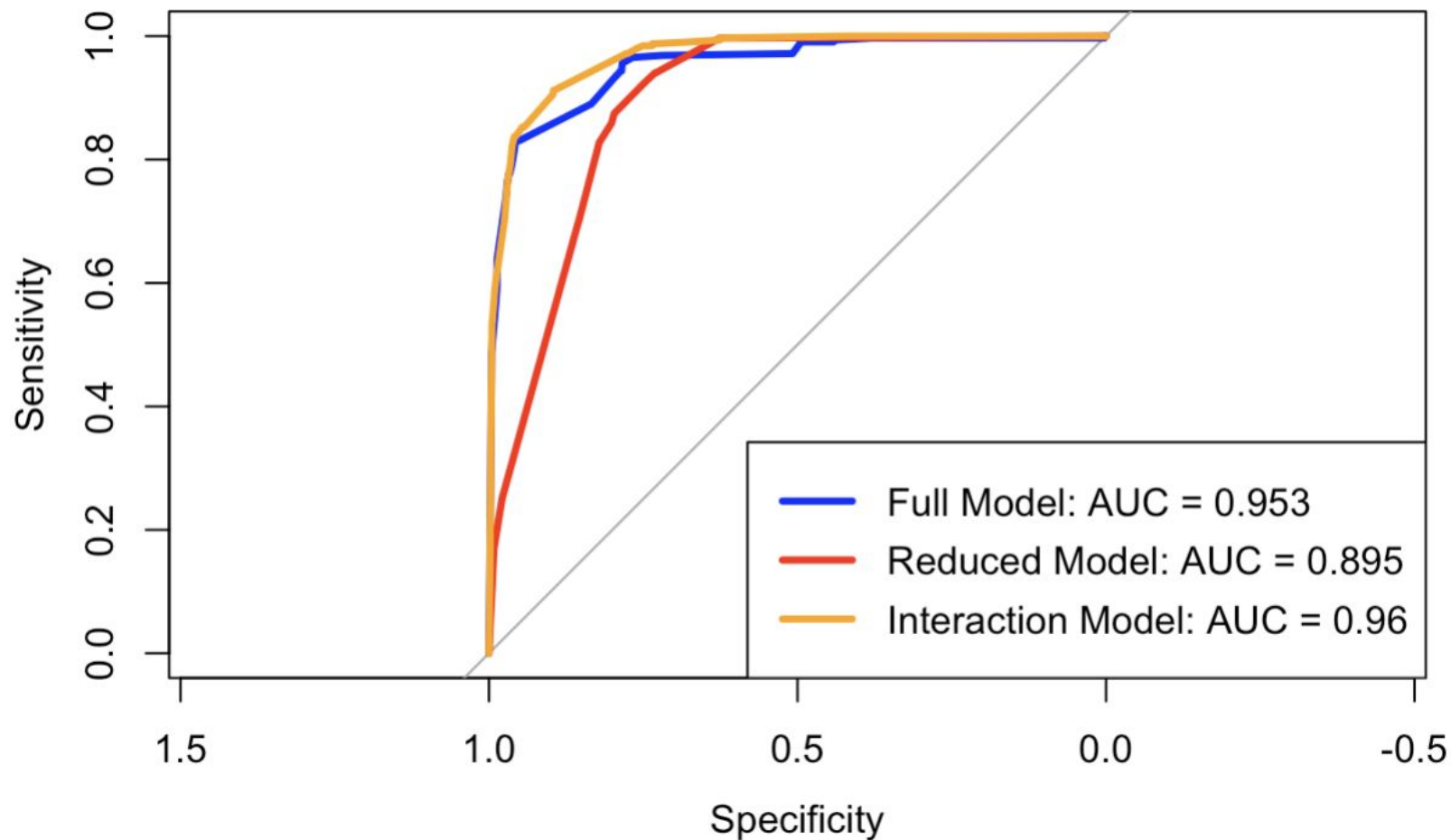
	OR	2.5%	97.5%
Intercept	0.054	0.039	0.074
Medium	0.591	0.474	0.737
Small	0.098	0.701	0.137
Iron	0.073	0.051	0.102
Stony-Iron	0.178	0.082	0.357
NE	83.272	62.839	111.547
NW	11.062	8.324	14.836
SW	17.028	11.086	25.945
Later	0.0121	0.009	0.015

Forest Plot: Factors Predicting 'Fell' Status

Adjusted Odds Ratios with 95% Confidence Intervals



ROC Curve Comparison



Confusion Matrix

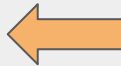
		Predicted	
Actual		Found	Fell
Found	8973	280	
Fell	75	244	



Full Model

- Sensitivity (Fell Detection) = 76.5%
- Specificity (Found Detection) = 97%

		Predicted	
Actual		Found	Fell
Found	9253	0	
Fell	319	0	



Reduced Model

- **Sensitivity = 0%**
- Specificity = 100%

		Predicted	
Actual		Found	Fell
Found	9206	47	
Fell	162	157	



Interaction Model

- Sensitivity = 49.2%
- Specificity = 99.4%

AIC

Model	Residual Deviance	AIC
Fall ~ 1	6531.7	6533.7
Fall ~ Mass	5153.8	5158.8
Fall ~ Composition	6524.8	6530.8
Fall ~ Region	5146.7	5154.7
Fall ~ Year	5567.1	5571.1
Fall ~ Mass + Region	4739.0	4751
Fall ~ Full Model	3105.6	3123.6
Fall ~ Interaction Model	2987.3	3017.3

Conclusion & Limitations

- Location (`quadrant`) seem to be the most important factor when determining whether meteor was observed falling or not
 - Meteors in the northern hemisphere have higher odds of being observed falling
 - Larger meteors also have higher odds of being observed falling
- The model with all terms seemed to perform the best in terms of classification
 - Note: Generally, more complex models are prone to overfitting
- Big class imbalances between found (~96%) and fell (~4%)

A composite image showing a volcanic eruption from a space perspective. A bright, glowing plume of smoke and ash rises from a rocky vent in the center, set against a backdrop of a starry night sky. Below the eruption, the curved horizon of the Earth is visible, showing blue oceans, white clouds, and green landmasses. The text "Thank you!" is centered in a bold, black, sans-serif font.

Thank you!